Enterprise Relational Database & Pipeline Specification

High-Throughput Normalized Data Storage & Ingestion Blueprint

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1. Executive Summary & Design Scope

Modern distributed commerce backends demand strict normalization across catalog, order, and logistics entities to eliminate anomalies while supporting elastic streaming workloads. This document formalizes the complete architectural blueprint and relational database design.

THROUGHPUT SLA	NORMALIZATION	REPLICATION LAG
10,000 QPS	**3NF Strict**	**< 250 ms**
99.9th percentile < 15ms	Zero transitive dependencies	Multi-region sync commit

2. Architectural Decomposition & Design Patterns

The processing pipeline decouples ingestion, validation, and multi-database persistence through three core design patterns:

Pattern	Component	Architectural Rationale
Strategy Pattern	Mode Hierarchy	Encapsulates mode execution algorithms (`DumpMode`, `PassthroughMode`, `ValidateMode`) behind unified interface.
Connector / Bridge	Persistence Subsystem	Isolates database drivers (PostgreSQL, Redis, Elasticsearch) from core processor logic.
Factory Pattern	ProcessorFactory	Centralizes dynamic instantiation and runtime dependency wiring during dynamic reconfiguration.

Critical Production Rule: Connection Pooling

Direct raw database connections crash under concurrent loads exceeding 100 parallel workers. All database connectors must utilize a shared connection pool (e.g. PgBouncer or HikariCP) configured with strict checkout timeouts.